Differentiation

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The chain rule**

Find $\frac{dy}{dx}$ for $y = (3x + 1)^5$.

$\frac{dy}{dx} = 5(3x + 1)^4 \cdot 3 = 15(3x + 1)^4$.

2 **[Foundational]** • **Target: The chain rule**

Find $\frac{dy}{dx}$ for $y = \sqrt{4x^2 + 1}$.

Write $y = (4x^2 + 1)^{1/2}$. $\frac{dy}{dx} = \frac{1}{2}(4x^2 + 1)^{-1/2} \cdot 8x = \frac{4x}{\sqrt{4x^2 + 1}}$.

3 **[Foundational]** • **Target: The chain rule**

Find $\frac{dy}{dx}$ for $y = (2x - 5)^6$.

$\frac{dy}{dx} = 6(2x - 5)^5 \cdot 2 = 12(2x - 5)^5$.

4 **[Foundational]** • **Target: The product rule**

Find $\frac{dy}{dx}$ for $y = x^2 \sin x$.

$\frac{dy}{dx} = 2x\sin x + x^2\cos x$.

5 **[Foundational]** • **Target: The product rule**

Find $\frac{dy}{dx}$ for $y = e^{3x}\cos x$.

$\frac{dy}{dx} = 3e^{3x}\cos x + e^{3x}(-\sin x) = e^{3x}(3\cos x - \sin x)$.

6 **[Foundational]** • **Target: The quotient rule**

Find $\frac{dy}{dx}$ for $y = \frac{x^2}{2x + 1}$.

$\frac{dy}{dx} = \frac{2x(2x + 1) - x^2(2)}{(2x + 1)^2} = \frac{4x^2 + 2x - 2x^2}{(2x + 1)^2} = \frac{2x^2 + 2x}{(2x + 1)^2} = \frac{2x(x + 1)}{(2x + 1)^2}$.

7 **[Foundational]** • **Target: The quotient rule**

Find $\frac{dy}{dx}$ for $y = \frac{\sin x}{x}$.

$\frac{dy}{dx} = \frac{x\cos x - \sin x}{x^2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: The quotient rule**

Find $\frac{dy}{dx}$ for $y = \frac{e^x}{x^2+1}$.

$\frac{dy}{dx} = \frac{e^x(x^2+1) - e^x(2x)}{(x^2+1)^2} = \frac{e^x(x^2-2x+1)}{(x^2+1)^2} = \frac{e^x(x-1)^2}{(x^2+1)^2}$.

9 **[Intermediate]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $x^2 + y^2 = 25$.

Differentiate both sides with respect to x: $2x + 2y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{x}{y}$.

10 **[Intermediate]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $x^3 + y^3 = 6xy$, and evaluate it at the point $(3, 3)$.

Differentiate: $3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$. Rearranging: $\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$, so $\frac{dy}{dx} = \frac{6y-3x^2}{3y^2-6x} = \frac{2y-x^2}{y^2-2x}$. At $(3, 3)$: $\frac{2(3)-9}{9-2(3)} = \frac{-3}{3} = -1$.

11 **[Intermediate]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $y^2 - xy + x^2 = 7$, and evaluate it at the point $(1, 3)$.

Differentiate: $2y\frac{dy}{dx} - \left(y + x\frac{dy}{dx}\right) + 2x = 0$. Rearranging: $\frac{dy}{dx}(2y - x) = y - 2x$, so $\frac{dy}{dx} = \frac{y-2x}{2y-x}$. At $(1, 3)$: $\frac{3-2}{6-1} = \frac{1}{5}$.

12 **[Intermediate]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $\sin(xy) = x$, using implicit differentiation.

Differentiate both sides: $\cos(xy)\left(y + x\frac{dy}{dx}\right) = 1$. Expanding: $y\cos(xy) + x\cos(xy)\frac{dy}{dx} = 1$. Rearranging: $\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy)}$.

13 **[Intermediate]** • **Target: Parametric differentiation**

A curve is defined parametrically by $x = t^2$, $y = 2t^3$. Find $\frac{dy}{dx}$ in terms of t.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6t^2$. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{6t^2}{2t} = 3t$ (for $t \neq 0$).

14 **[Intermediate]** • **Target: Parametric differentiation**

A curve is defined by $x = 3\cos t$, $y = 3\sin t$. Find $\frac{dy}{dx}$ in terms of t, and find the gradient at $t = \frac{\pi}{4}$.

$\frac{dx}{dt} = -3\sin t$, $\frac{dy}{dt} = 3\cos t$. $\frac{dy}{dx} = \frac{3\cos t}{-3\sin t} = -\cot t$. At $t = \frac{\pi}{4}$: $-\cot\left(\frac{\pi}{4}\right) = -1$, since $\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Related rates • Cross-Lesson: 5.01 Integration

A spherical balloon is inflated so its volume increases at a constant rate of 100 cm^3/s. Find the rate of increase of the radius when $r = 5$ cm, using $V = \frac{4}{3}\pi r^3$, to three decimal places.

$\frac{dV}{dr} = 4\pi r^2$. At $r = 5$: $\frac{dV}{dr} = 4\pi(25) = 100\pi$. $\frac{dr}{dt} = \frac{dV/dt}{dV/dr} = \frac{100}{100\pi} = \frac{1}{\pi} \approx 0.318$ cm/s.

16 [Challenging] • Target: Related rates • Cross-Lesson: 6.01 Sequences & Series

The area of a circular ripple increases at 8π cm^2/s. Find the rate of change of the radius when $r = 4$ cm, using $A = \pi r^2$.

$\frac{dA}{dr} = 2\pi r$. At $r = 4$: $\frac{dA}{dr} = 8\pi$. $\frac{dr}{dt} = \frac{dA/dt}{dA/dr} = \frac{8\pi}{8\pi} = 1$ cm/s.

17 [Challenging] • Target: Related rates • Cross-Lesson: 7.01 Vectors

A cone-shaped tank (vertex down) has height 10 cm and top radius 4 cm, so at water depth h the radius is $r = \frac{2h}{5}$. Water is poured in at 12 cm^3/s. Find the rate of rise of the water level when $h = 5$ cm, using $V = \frac{1}{3}\pi r^2 h$.

Substitute $r = \frac{2h}{5}$: $V = \frac{1}{3}\pi\left(\frac{2h}{5}\right)^2 h = \frac{4\pi}{75}h^3$. $\frac{dV}{dh} = \frac{12\pi}{75}h^2 = \frac{4\pi}{25}h^2$. At $h = 5$: $\frac{dV}{dh} = \frac{4\pi}{25}(25) = 4\pi$. $\frac{dh}{dt} = \frac{dV/dt}{dV/dh} = \frac{12}{4\pi} = \frac{3}{\pi} \approx 0.955$ cm/s.

18 [Challenging] • Target: Related rates • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

The radius of a circular oil slick increases at a constant rate of 0.5 m/min. Find the rate of increase of its area when $r = 20$ m.

$A = \pi r^2$, so $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$. At $r = 20$: $\frac{dA}{dt} = 2\pi(20)(0.5) = 20\pi \approx 62.83$ m^2/min.

19 [Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 9.01 Complex Numbers

Find $\frac{dy}{dx}$ for $y = \sin(3x^2)$.

$\frac{dy}{dx} = \cos(3x^2) \cdot 6x = 6x\cos(3x^2)$.

20 [Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find $\frac{dy}{dx}$ for $y = \ln(5x^2 + 1)$.

$\frac{dy}{dx} = \frac{10x}{5x^2 + 1}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Product & Quotient Rules

Differentiate $y = x^3 e^{2x}$ with respect to x.

a. $x^2e^{2x}(3+2x)$ ✓

b. $3x^2e^{2x}$

c. $2x^3e^{2x}$

d. $x^2e^{2x}(3x+2)$

Product rule: $dy/dx = (3x^2)e^{2x} + x^3(2e^{2x}) = x^2e^{2x}(3+2x)$.

22 [Foundational] • Target: Product & Quotient Rules

Differentiate $y = \frac{\ln x}{x^2}$ for $x > 0$.

a. $\frac{1-2\ln x}{x^3}$ ✓

b. $\frac{1+2\ln x}{x^3}$

c. $\frac{1-2\ln x}{x^4}$

d. $\frac{1}{x^3}$

Quotient rule: $\frac{(1/x)(x^2)-(\ln x)(2x)}{x^4} = \frac{x-2x\ln x}{x^4} = \frac{1-2\ln x}{x^3}$.

23 [Foundational] • Target: Product & Quotient Rules

Find the gradient of the curve $y = x\sin(2x)$ at $x = \pi/4$.

a. **1** ✓

b. 0

c. $\pi/2$

d. 2

$dy/dx = \sin(2x) + 2x\cos(2x)$. At $x = \pi/4$: $\sin(\pi/2) + 2(\pi/4)\cos(\pi/2) = 1 + (\pi/2)(0) = 1$.

24 [Foundational] • Target: Derivatives of Exponential, Logarithmic & Trig Functions

Differentiate $f(x) = \ln(\sec x + \tan x)$.

a. $\sec x$ ✓

b. $\tan x$

c. $\sec^2 x$

d. $\sec x \tan x$

$f'(x) = \frac{\sec x\tan x + \sec^2 x}{\sec x + \tan x} = \frac{\sec x(\tan x + \sec x)}{\sec x + \tan x} = \sec x$.

25 [Foundational] • Target: Derivatives of Exponential, Logarithmic & Trig Functions

Differentiate $y = e^{\sin(3x)}$ with respect to x.

a. $3\cos(3x)e^{\sin(3x)}$ ✓

b. $\cos(3x)e^{\sin(3x)}$

c. $3e^{\sin(3x)}$

d. $-3\cos(3x)e^{\sin(3x)}$

Chain rule: $d/dx[e^{\sin(3x)}] = e^{\sin(3x)} \cdot d/dx[\sin(3x)] = 3\cos(3x)e^{\sin(3x)}$.

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26 [Foundational] • Target: Derivatives of Exponential, Logarithmic & Trig Functions

Find $\frac{d}{dx}[\tan^{-1}(2x)]$.

a. $\frac{2}{1+4x^2}$ ✓

b. $\frac{1}{1+4x^2}$

c. $\frac{2}{1+2x^2}$

d. $\frac{2}{\sqrt{1-4x^2}}$

$\frac{d}{dx}[\arctan(u)] = \frac{u'}{1+u^2} = \frac{2}{1+(2x)^2} = \frac{2}{1+4x^2}$.

27 [Foundational] • Target: Implicit Differentiation

Find $\frac{dy}{dx}$ for the curve $x^2 + xy + y^2 = 7$.

a. $-\frac{2x+y}{x+2y}$ ✓

b. $\frac{2x+y}{x+2y}$

c. $-\frac{2x}{2y}$

d. $-\frac{x+2y}{2x+y}$

$2x + (y + x dy/dx) + 2y dy/dx = 0 \Rightarrow dy/dx(x+2y) = -(2x+y) \Rightarrow dy/dx = -\frac{2x+y}{x+2y}$.

28 [Foundational] • Target: Implicit Differentiation

Find the gradient of the curve $x^3 + y^3 = 9xy$ at the point $(2, 4)$.

a. $\frac{4}{5}$ ✓

b. $-\frac{4}{5}$

c. $\frac{5}{4}$

d. 1

$3x^2 + 3y^2y' = 9(y + xy') \Rightarrow x^2 + y^2y' = 3y + 3xy'$. At $(2, 4)$: $4 + 16y' = 12 + 6y' \Rightarrow 10y' = 8 \Rightarrow y' = 8/10 = 4/5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Implicit Differentiation

Find the equation of the normal to the curve $x^2 - 2xy + 3y^2 = 9$ at $(3, 2)$.

a. $y = 2$ ✓

b. $x = 3$

c. $y = x - 1$

d. $2x + y = 8$

Differentiate: $2x - 2(y + xy') + 6yy' = 0$. At $(3, 2)$: $6 - 2(2 + 3y') + 12y' = 0 \Rightarrow 6 - 4 - 6y' + 12y' = 0 \Rightarrow 2 + 6y' = 0 \Rightarrow y' = -1/3$.
Normal slope is 3. Equation: $y - 2 = 3(x - 3) \Rightarrow y = 3x - 7$.

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30 [Intermediate] • Target: Parametric Differentiation

A curve is defined parametrically by $x = 2t^2$, $y = 4t$. Find $\frac{dy}{dx}$ in terms of t.

a. $\frac{1}{t}$ ✓

b. t

c. $\frac{2}{t}$

d. $2t$

$dx/dt = 4t$, $dy/dt = 4$. Then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{4}{4t} = \frac{1}{t}$.

31 [Intermediate] • Target: Parametric Differentiation

A curve has parametric equations $x = a\cos\theta$, $y = b\sin\theta$. Find the equation of the tangent at parameter θ.

a. $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$ ✓

b. $\frac{x}{a}\sin\theta + \frac{y}{b}\cos\theta = 1$

c. $\frac{x}{a}\cos\theta - \frac{y}{b}\sin\theta = 1$

d. $bx\cos\theta + ay\sin\theta = ab$

$dx/d\theta = -a\sin\theta$, $dy/d\theta = b\cos\theta \Rightarrow dy/dx = -\frac{b\cos\theta}{a\sin\theta}$. Tangent: $y - b\sin\theta = -\frac{b\cos\theta}{a\sin\theta}(x - a\cos\theta) \Rightarrow \frac{y\sin\theta}{b} - \sin^2\theta = -\frac{x\cos\theta}{a} + \cos^2\theta \Rightarrow \frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$.

32 [Intermediate] • Target: Parametric Differentiation

Find $\frac{d^2y}{dx^2}$ for the parametric curve $x = t^2$, $y = t^3$.

a. $\frac{3}{4t}$ ✓

b. $\frac{3}{2}$

c. $\frac{3}{4t^2}$

d. $\frac{6}{t}$

$dx/dt = 2t$, $dy/dt = 3t^2 \Rightarrow dy/dx = 3t/2$. $\frac{d^2y}{dx^2} = \frac{d}{dt}[dy/dx]/\frac{dx}{dt} = \frac{3/2}{2t} = \frac{3}{4t}$.

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33 [Intermediate] • Target: Parametric Differentiation

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$ ✓

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

We synthesize differentiation with coordinate geometry to find the two points of contact. **Step 1: Point of tangency on $C_2 : y = \ln x$** Let $A(x_1, y_1)$ be the point of contact on C_2, so $y_1 = \ln x_1$. The derivative is $\frac{dy}{dx} = \frac{1}{x}$, so the gradient of the tangent at A is $\frac{1}{x_1}$. Since the tangent line passes through the origin $(0, 0)$, its gradient is also: $m = \frac{y_1 - 0}{x_1 - 0} = \frac{\ln x_1}{x_1}$ Equating the two expressions for the gradient.

34 [Intermediate] • Target: The chain rule

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

b. 45 ✓

c. 50

d. 54

Let the arithmetic progression (u_n) have first term a and common difference d. Then: $u_1 = a, \quad u_4 = a + 3d, \quad u_6 = a + 5d.$ Let the geometric progression (v_n) have first term v_1 and common ratio r. Since $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$, we have: $v_1 = a, \quad v_2 = ar, \quad v_3 = ar^2.$ Equating terms: 1) $a + 3d = ar \Rightarrow 3d = a(r-1) \Rightarrow d = \frac{a(r-1)}{3}$. 2) $a + 5d = ar^2$. Substitute d into the second equation.

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35 [Intermediate] • Target: The product rule

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$ ✓

b. $k = -2;\ x = 7$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

1. **Find the parameter k:** By the definition of an inverse function, $f^{-1}(4) = 2 \Leftrightarrow f(2) = 4$. Substituting $x = 2$ into $f(x)$: $f(2) = 3(2) + k = 6 + k\ 6 + k = 4 \Rightarrow k = -2$ 2. **Determine the inverse function $f^{-1}(x)$:** With $k = -2$, $f(x) = 3x - 2$. Setting $y = 3x - 2$ and solving for x: $3x = y + 2 \Rightarrow x = \frac{y+2}{3}$ Thus, $f^{-1}(x) = \frac{x+2}{3}$. 3. **Solve the equation $f^{-1}(x) = g(x)$:** $\frac{x+2}{3} = |x - 4| \Leftrightarrow x + 2 = 3|x - 4|$.

36 [Intermediate] • Target: The quotient rule

The curve C has equation $y = \frac{ax+6}{x^2+3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

a. $(-3, -1)$ ✓

b. $(-3, 1)$

c. $(3, 1)$

d. $(-3, -2)$

To find the unknown parameter a, differentiate y with respect to x using the quotient rule: $\frac{dy}{dx} = \frac{a(x^2+3) - (ax+6)(2x)}{(x^2+3)^2} = \frac{3a - 12x - ax^2}{(x^2+3)^2}$ At a stationary point, $\frac{dy}{dx} = 0$, which requires the numerator to be zero: $3a - 12x - ax^2 = 0$ Given that $x = 1$ is a stationary point: $3a - 12(1) - a(1)^2 = 0 \Rightarrow 2a - 12 = 0 \Rightarrow a = 6$ Substitute $a = 6$ back into the numerator of the derivative.

37 [Intermediate] • Target: Implicit differentiation

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

a. 2 ✓

b. 6

c. $\frac{2}{3}$

d. $\ln 3$

To find the gradient of the tangent to the curve $y = \ln(2x^3 + 1)$ at $x = 1$, we differentiate y with respect to x using the chain rule. Let $u = 2x^3 + 1$, so $y = \ln(u)$. Then: $\frac{du}{dx} = 6x^2$ and $\frac{dy}{du} = \frac{1}{u}$ Applying the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{6x^2}{2x^3+1}$ Now evaluate the derivative at $x = 1$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Parametric differentiation** • **Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

What are the values of x that satisfy the equation $|3x-2| = 2|x+1|$?

a. $x = 0$ **and** $x = 4$ ✓

b. $x = 0$ and $x = -4$

c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

To solve the modulus equation $|3x-2| = 2|x+1|$, we can either consider the linear cases or square both sides. **Method 1: Considering linear cases** The equation $|A| = |B|$ (where $B = 2(x+1)$) is equivalent to $A = B$ or $A = -B$: - Case 1: $3x-2 = 2(x+1) \Rightarrow 3x-2 = 2x+2 \Rightarrow x = 4$ - Case 2: $3x-2 = -2(x+1) \Rightarrow 3x-2 = -2x-2 \Rightarrow 5x = 0 \Rightarrow x = 0$ **Method 2: Squaring both sides** Since both sides are non-negative, we can square both sides.

39 **[Challenging]** • **Target: Related rates** • **Cross-Lesson: 9.01 Complex Numbers**

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$ ✓

b. $6e^3$

c. $3e^3$

d. $2e^3$

To find the gradient of the curve, differentiate $y = x^2e^{3x}$ with respect to x using the product rule: $\frac{d}{dx}[uv] = u\frac{dv}{dx} + v\frac{du}{dx}$ Let $u = x^2$ and $v = e^{3x}$. Then $\frac{du}{dx} = 2x$ and, by the chain rule, $\frac{dv}{dx} = 3e^{3x}$. Applying the product rule: $\frac{dy}{dx} = x^2(3e^{3x}) + 2x(e^{3x}) = (3x^2 + 2x)e^{3x}$ Substitute $x = 1$.

40 **[Challenging]** • **Target: Differentiating trigonometric, exponential and logarithmic composites** • **Cross-Lesson: 1.01 Functions: Modulus & Inverse**

Given $f(x) = -3x + 4$ and $g(x) = -4x - 6$, find $gf(3)$.

a. **14** ✓

b. 58

c. -18

d. -5

Applying Functions: Modulus & Inverse principles, $f(3) = -5$, so $gf(3) = g(-5) = 14$. Therefore, the verified correct option is 14.

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41 [Challenging] • Target: Applying the rules: tangents and second derivatives • Cross-Lesson: 10.01 Numerical Methods

The function $f(x) = (x + 7)^2 - 15$ has domain $x \geq -7$. State the range of f.

a. $f(x) \geq -7$

b. $x \geq -7$

c. $f(x) \leq -15$

d. $f(x) \geq -15$ ✓

Since $x \geq -7$, the minimum of f occurs at $x = -7$, giving $f(-7) = -15$, and f increases thereafter. Range: $f(x) \geq -15$.

42 [Challenging] • Target: A well-designed word problem: a ladder sliding down a wall • Cross-Lesson: 11.01 Differential Equations

Given $f(x) = -5x - 18$, evaluate $f(|0|)$.

a. -15

b. **-18** ✓

c. -14

d. 18

Applying Functions: Modulus & Inverse principles, $|0| = 0$, so $f(|0|) = f(0) = -18$. Therefore, the verified correct option is -18.

43 [Challenging] • Target: Product & Quotient Rules • Cross-Lesson: 12.01 Poisson Distribution

Given $f(x) = -x - 10$ and $g(x) = -3x + 9$, find $fg(6)$.

a. **-1** ✓

b. 57

c. -9

d. -16

Applying Functions: Modulus & Inverse principles, $g(6) = -9$, so $fg(6) = f(-9) = -1$. Therefore, the verified correct option is -1.

44 [Challenging] • Target: Derivatives of Exponential, Logarithmic & Trig Functions • Cross-Lesson: 13.01 Continuous Random Variables

The function $f(x) = (x + 11)^2 + 5$ has domain $x \geq -11$. State the range of f.

a. $x \geq -11$

b. $f(x) \geq -11$

c. $f(x) \leq 5$

d. $f(x) \geq 5$ ✓

Since $x \geq -11$, the minimum of f occurs at $x = -11$, giving $f(-11) = 5$, and f increases thereafter. Range: $f(x) \geq 5$.

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45 [Challenging] • Target: Implicit Differentiation • Cross-Lesson: 14.01 Sampling & Estimation

The function $f(x) = (x + 6)^2 - 3$ has domain $x \geq -6$. State the range of f.

a. $f(x) \leq -3$

b. $f(x) \geq -3$ ✓

c. $f(x) \geq -6$

d. $x \geq -6$

Since $x \geq -6$, the minimum of f occurs at $x = -6$, giving $f(-6) = -3$, and f increases thereafter. Range: $f(x) \geq -3$.

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